

Lab 3: Component Modelling & Architectural Pattern Selection

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Architecture Selection

We chose Layered Architecture for the Patient Health Record Consent Management System.

Architectural Choice

A three-layer style (Presentation, Business, Data) was selected. The User Interface component sits in the Presentation layer; the Consent Manager, Doctor Verification, Notification and Audit Trail components sit in the Business layer; and the Database component sits in the Data layer. The components communicate only through the interfaces IConsentService, IDoctorVerification, INotification, IAuditLog, IConsentStore and IAuditStore. Microservices were rejected because their network latency, operational complexity and data-consistency challenges outweigh their benefits for a system of this size. A plain Client-Server design was rejected because it does not separate business rules from data and leaves a single point of failure.

Two Reasons for the Choice

1. **Direct mapping to requirements and easy replacement.** Each Lab 1 requirement lands in one component: FR-001, FR-004 and FR-005 (grant, request, revoke) in the Consent Manager, FR-002 (revocation confirmation) in the Notification component, and NFR-001 (audit trail) in the Audit Trail component. Because every component hides its implementation behind an interface, an email/SMS provider or the database can be swapped without touching other layers. This also suits the small team and one-week sprints from Lab 2.
2. **Consistency of consent state.** A revocation must take effect and be logged together, and a clinic must never read records after expiry (FR-001 and FR-005 fail conditions). One shared Data layer allows consent changes and audit entries to be written in a single transaction. Microservices would need distributed transactions or eventual consistency, which risks a stale-consent window.

Security Advantage

All access flows through a single enforcement point. The UI never reaches the Data layer directly; every grant, request and record access passes through the Consent Manager, which calls Doctor Verification first and then checks the expiry timestamp. An unverified doctor therefore cannot bypass the check (FR-004, Alternate Flow). The Audit Trail is reached only through IAuditStore, which exposes an INSERT-only operation, so audit records cannot be edited or deleted (NFR-001). The Data layer is not exposed to clients, which supports the privacy requirement NFR-002.

Performance Benefit

Calls between layers are in-process method or API calls rather than network hops, so the consent check that runs on every record access has low latency. The Consent Manager can cache active consents, and the Notification component can be invoked asynchronously so that sending the email/SMS does not delay revocation. The main drawback of layering is that individual components are harder to scale, but the stateless Business layer can be replicated behind a load balancer if load increases.